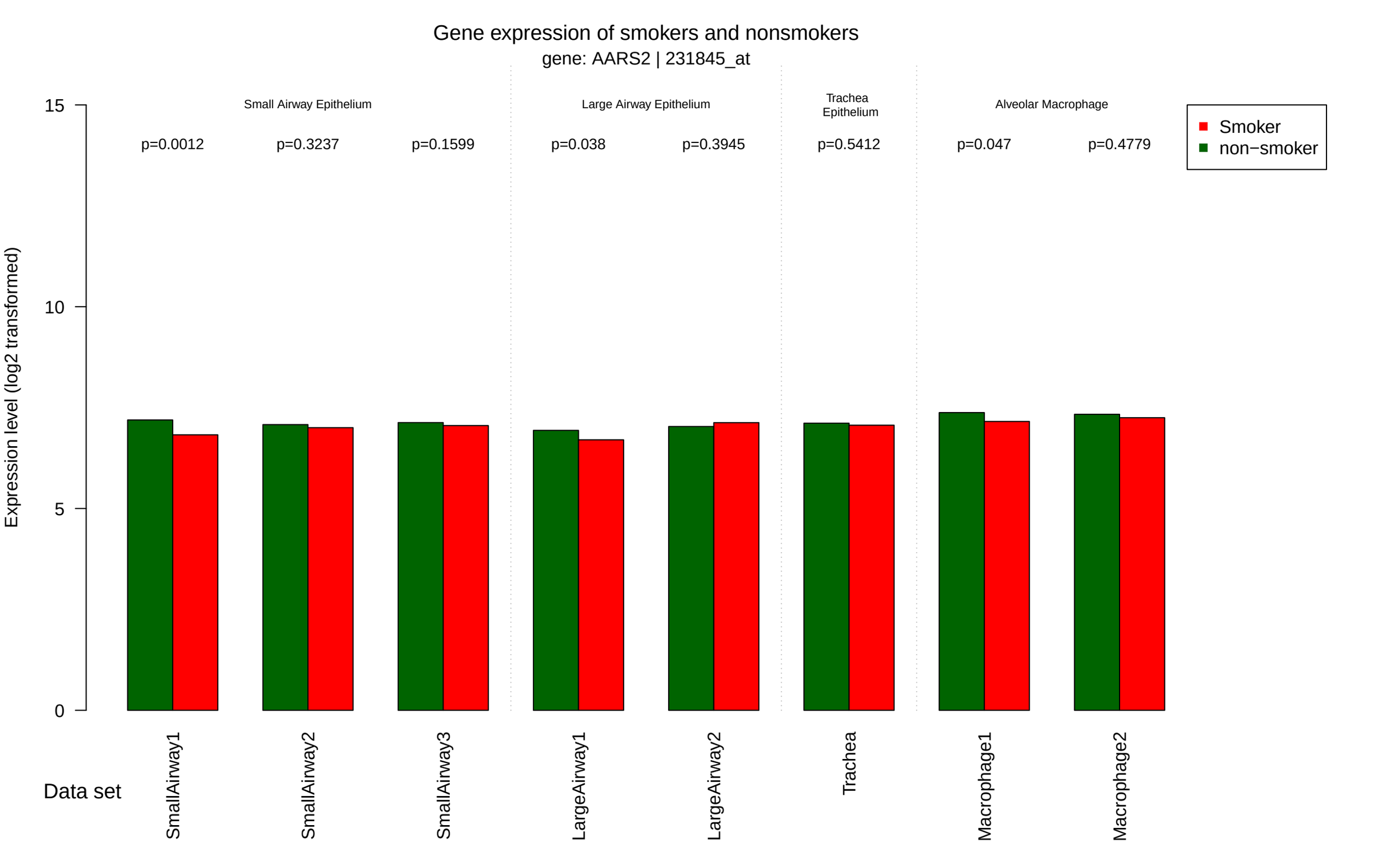
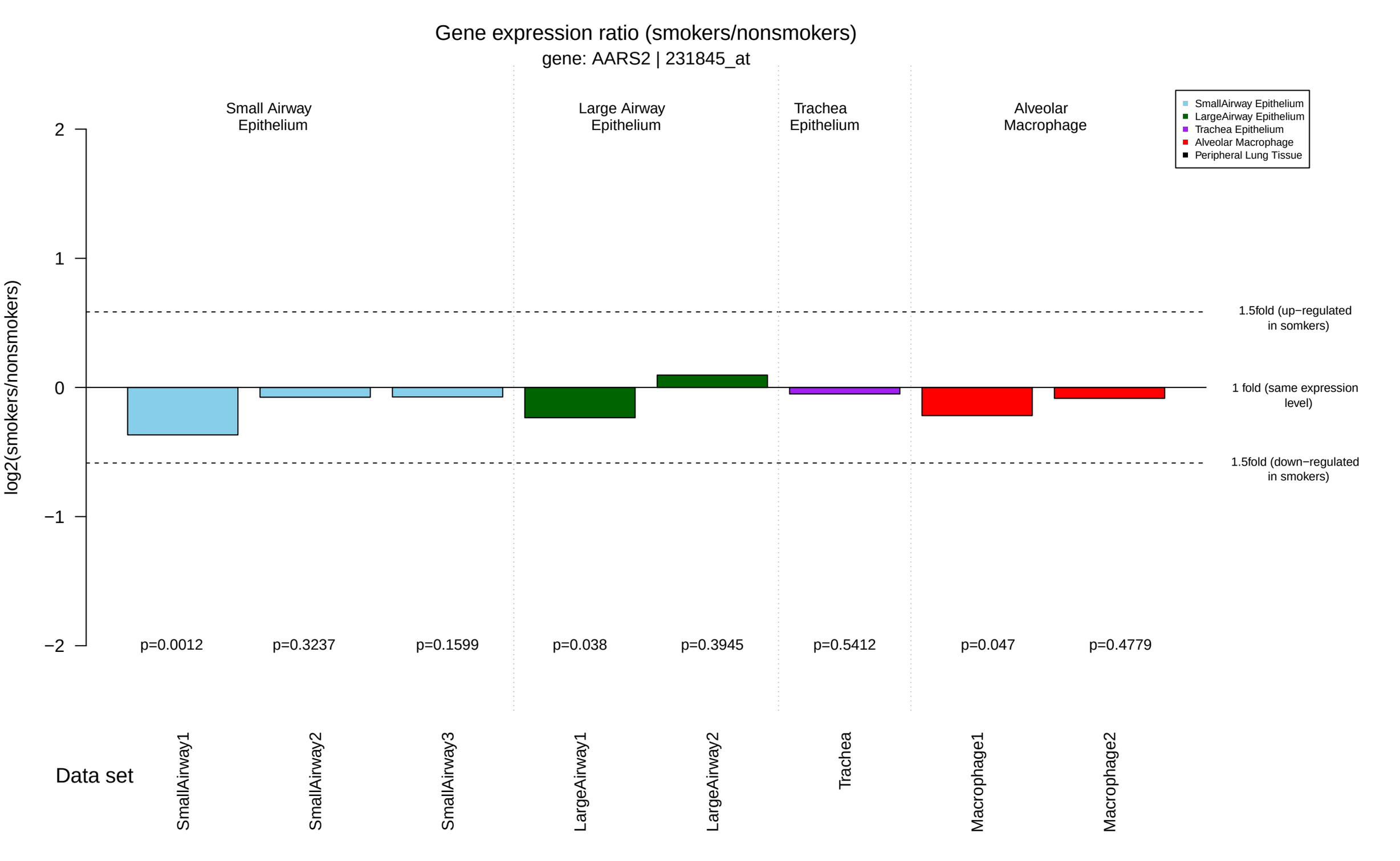
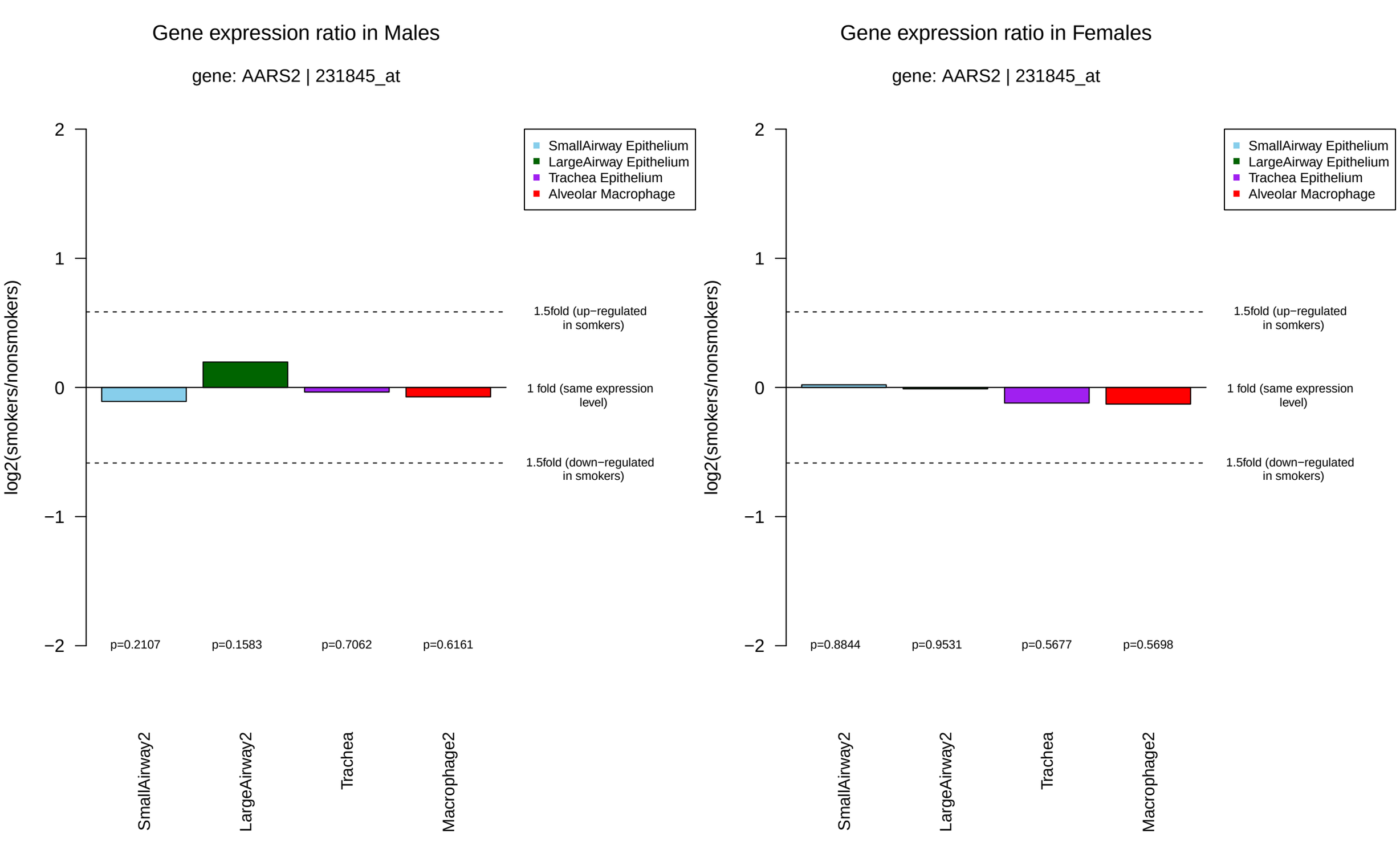
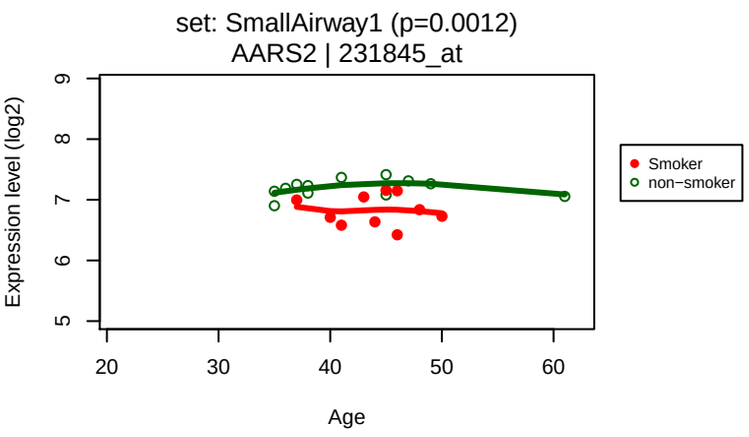
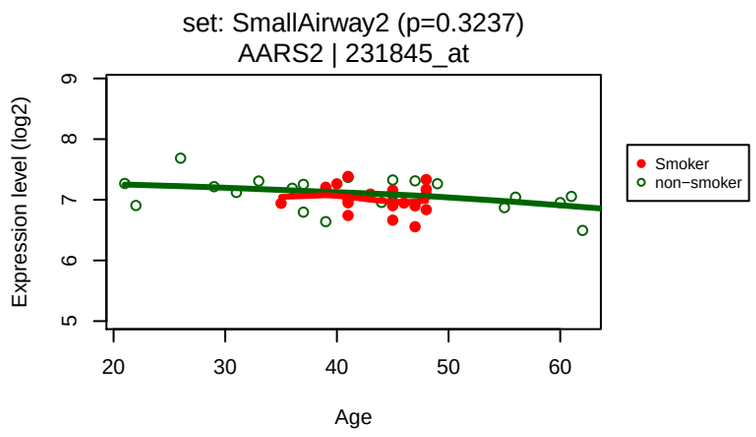
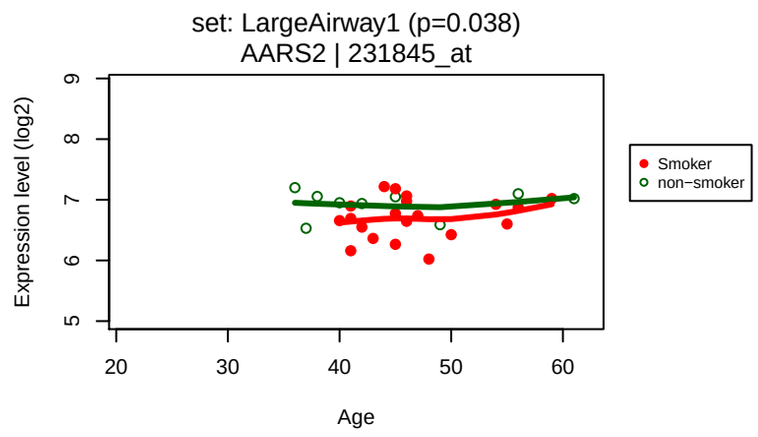

Gene expression of smokers and nonsmokers
gene: AARS2 | 231845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0012
p=0.3237
p=0.1599
p=0.038
p=0.3945
p=0.5412
p=0.047
p=0.4779
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

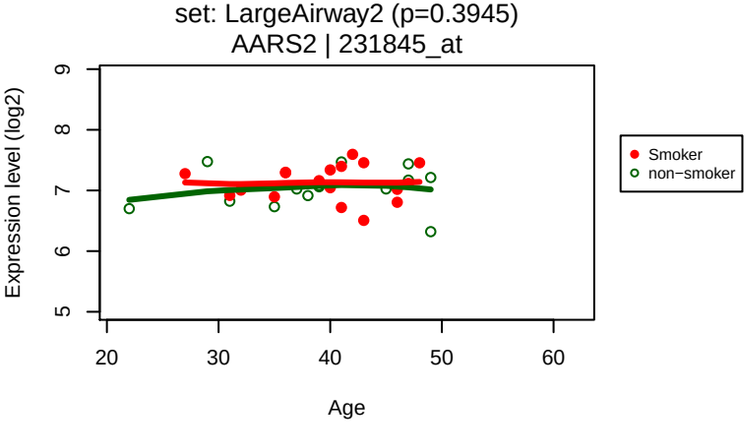

Gene expression ratio (smokers/nonsmokers)
gene: AARS2 | 231845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.3237
p=0.1599
p=0.038
p=0.3945
p=0.5412
p=0.047
p=0.4779
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

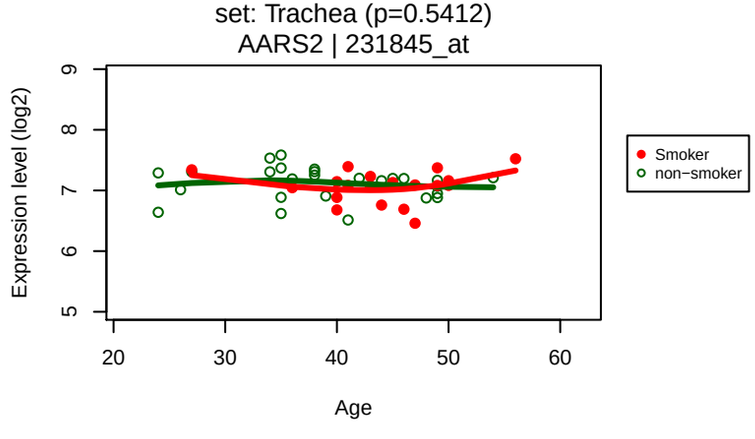

Gene expression ratio in Males
gene: AARS2 | 231845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2107
p=0.1583
p=0.7062
p=0.6161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AARS2 | 231845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8844
p=0.9531
p=0.5677
p=0.5698
SmallAirway2
LargeAirway2
Trachea
Macrophage2

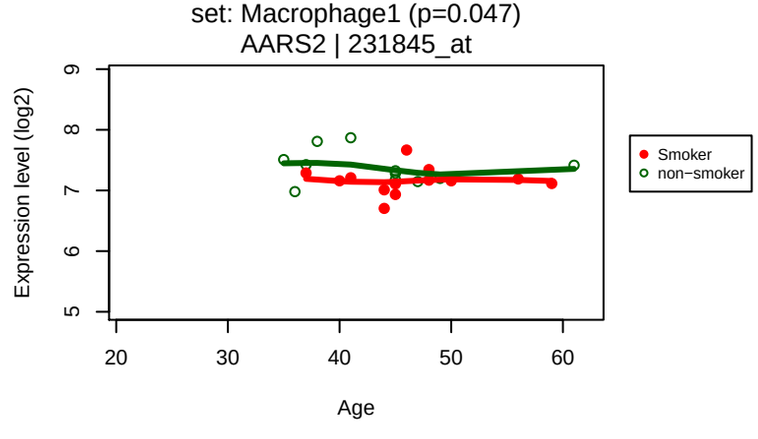

set: SmallAirway1 (p=0.0012)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker

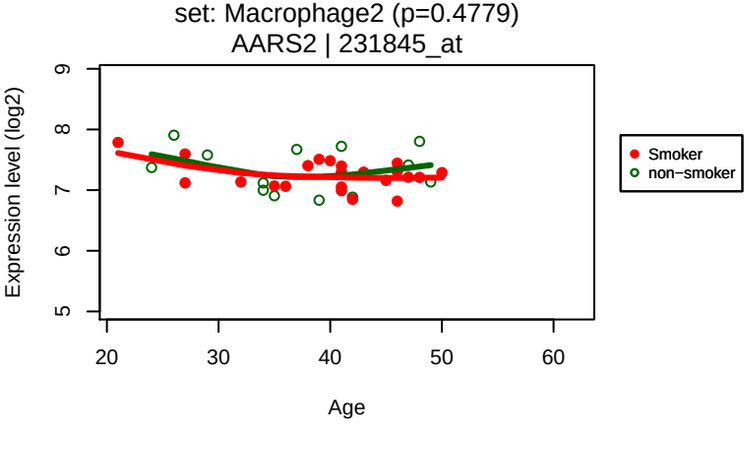

set: SmallAirway2 (p=0.3237)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker

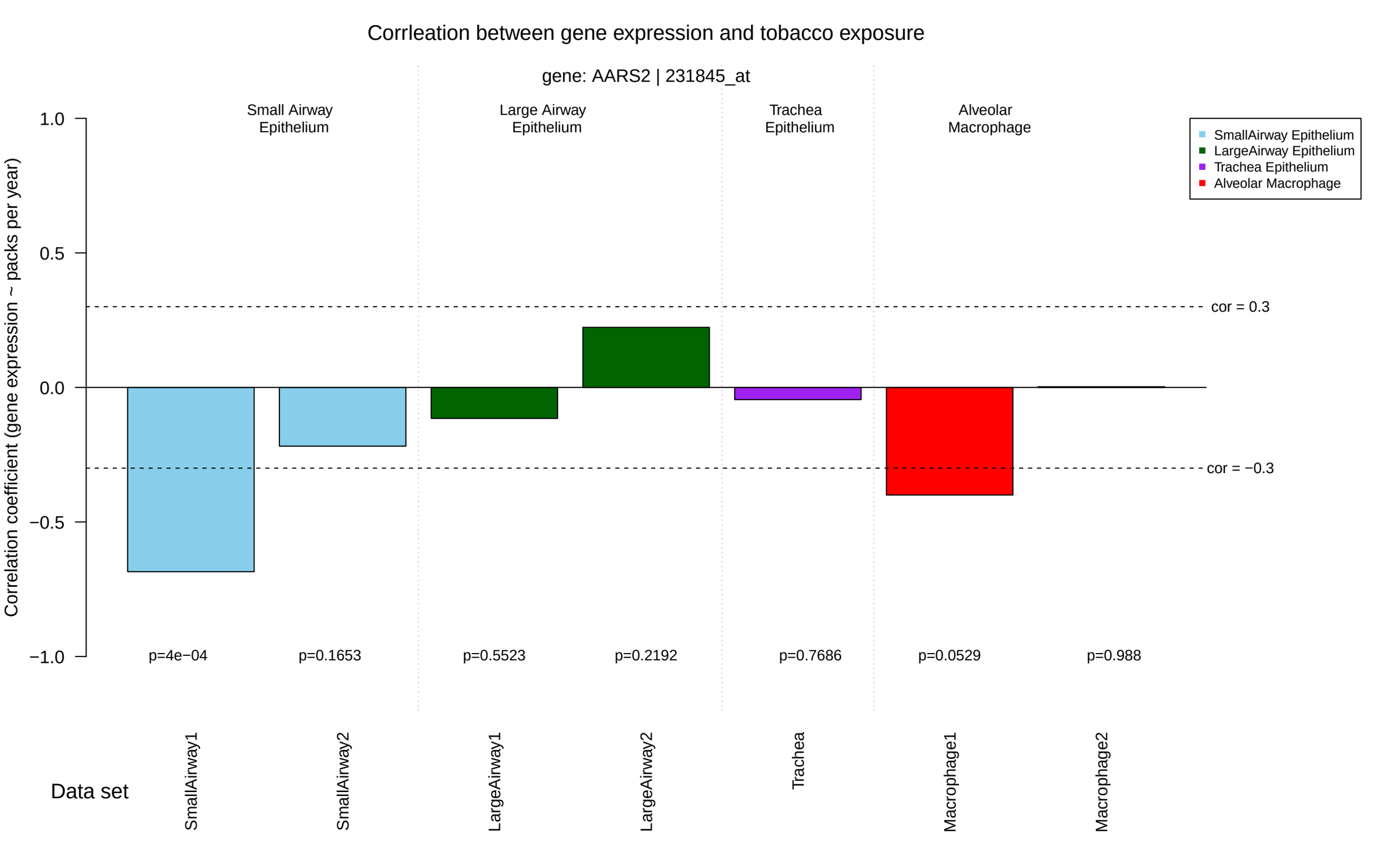

set: LargeAirway1 (p=0.038)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3945)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5412)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.047)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4779)
AARS2 | 231845_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AARS2 | 231845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.1653
p=0.5523
p=0.2192
p=0.7686
p=0.0529
p=0.988
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2